MTM4692

Spring 2026

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Key Information

- Monday, 1:00 pm - 3:50 pm @ online.yildiz.edu.tr
- Bologna : http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=9563&aid=24&pid=37
- Grade-Thresholds: http://www.bologna.yildiz.edu.tr/index.php?r=institution/grading
- Attendance: Required by **REGULATIONS!!!**

Grading Breakdown (100%)

Mid-Term(30%) and Final Exam (40%)

- Face to Face. Room? : To be announced.

Project (30%)

- Needs to solve a real world problems like at YTU Bologna. Contribution?
- Number of team members is 3 to 5. Why?
  - **Project Milestone (10%):** Share and justify your idea why it's important in a technical report.
  - **Project Final (20%):** Presentation and project files.
- **Recap Quizes or Questions** during the class: Extra up to 10% and will be added your final project.

Semester Project

- **Team size:** 3–5 members
- **Team submission due:** March 9, 23:00 PM
- **Milestone (10%):** Technical report — share & justify your idea
- **Final (20%):** Presentation + project files
- You will build a **complete database application** using SQLite
- Each week's coding practice prepares you for the project

Sample Project Ddeas:

`University Course Registration & GPA Tracker, Hospital Appointment & Patient Management System, E-Commerce Order & Inventory Management System`

Lecture 1 - Basic Concepts

SQLite Concepts and Environment Setup

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Overview

1. Introduction & Icebreaker
2. Environmental Setup (VSC + SQLite)
3. The Transition: Spreadsheets to Databases
4. SQLite & VS Code Integration
5. Basic Querying (SELECT, LIMIT)
6. Filtering Data (WHERE, LIKE, Ranges)
7. Data Organization (ORDER BY)
8. Aggregate Functions (COUNT, AVG, SUM)

1. Introduction & Icebreaker

First Contact with Data

- **Question:** Which database system might you have used, knowingly or unknowingly, since you woke up today?
- **Examples:** Social media feeds, Banking apps, YouTube, or the YTU Student Information System.

Why SQL for Math Engineers?

- **Scalability:** Handling millions of rows where Excel crashes.
- **Declarative Logic:** You describe *what* you want, not *how* to get it.
- **Relational Algebra:** Implementation of Set Theory (Unions, Intersections).
- **AI Pipeline:** SQL is the first step in any Machine Learning workflow.

2. Tables vs. Spreadsheets

Feature	Spreadsheet (Excel)	Database (SQL)
User Load	One user at a time	Thousands of users
Performance	Drops with file size	Scalable to Terabytes
Data Types	Flexible/Messy	Strict and Defined
History	1980s onwards	Ancient (Table Concept)

What Is a Database?

A **database** is an organized collection of structured information or data, typically stored electronically.

Era	Storage Method
1960s	Hierarchical & network databases
1970s	**Relational model** (E.F. Codd)
1980s–90s	Commercial RDBMS (Oracle, SQL Server)
2000s	NoSQL movement (MongoDB, Cassandra)
2010s+	Cloud databases, NewSQL

> **Key goal:** Persistent, reliable, and efficient data storage & retrieval

4. Environment Setup: The Stack

We use **VS Code** + **SQLite**.

- **Why?** SQLite is a C-library that provides a lightweight disk-based database.
- No server process needed.
- Perfect for edge computing and mobile apps.

Installation: Windows 🪟

1. **VSC:** Install from `code.visualstudio.com`.
2. **SQLite:** Extract `sqlite-tools` to `C:\sqlite`.
3. **Environment:** Add `C:\sqlite` to your System Path.
4. **Extension:** Install "SQLite" by *alexcvzz*.

Installation: macOS & Linux 🍎🐧

1. **Terminal Check:** `sqlite3 --version`.
2. **Install:** `brew install sqlite` or `sudo apt install sqlite3`.
3. **Extension:** Search for "SQLite" in VSC Marketplace.

3. Anatomy of a Table

- **Fields (Columns):** Typed attributes (Integer, Text, Real).
- **Records (Rows):** Instances of the entity.
- **Primary Key:** A unique identifier for every row.

Relational Algebra Basics

SQL is based on **Tuple Relational Calculus**.

- **Selection (σ):** Filtering rows (WHERE).
- **Projection (π):** Choosing columns (SELECT).

5. Practice Session

Git Repo: Github: https://github.com/f-kiran/mtm4692-sql-practice

Open your terminal in VSC and load the database:

```
sqlite3 longlist.db
```

This loads our dataset containing 100+ years of literary awards.

Dot Commands

These are specific to the SQLite CLI, not standard SQL:

- `.mode box`: Prettify output.
- `.schema`: View table structures.
- `.tables`: List all tables.
- `.quit`: Exit.

SELECT: The Foundation

The `SELECT` statement determines which columns to display.

```
SELECT title, author FROM longlist;
```

SELECT ALL (*)

The asterisk `*` acts as a wildcard for "all columns."

- *Warning:* Avoid using `*` in production code to save bandwidth and memory.

```
SELECT * FROM longlist;
```

LIMITing the Firehose

Databases can hold billions of rows. Never `SELECT *` without a `LIMIT` during exploration.

```
SELECT title FROM longlist LIMIT 5;
```

WHERE: The Filter

The `WHERE` clause defines the search condition.

```
SELECT title FROM longlist WHERE year = 2023;
```

Working with Strings

Strings in SQL must be enclosed in **single quotes**.

```
SELECT * FROM longlist WHERE author = 'Toni Morrison';
```

Comparison Operators

- `=` : Equal
- `!=` or `<>` : Not Equal
- `<` / `>` : Less/Greater than
- `<=` / `>=` : Inclusive bounds

Logic Gates: AND

Find records meeting **multiple** criteria.

```
SELECT title FROM longlist
WHERE year = 2022 AND rating > 4.5;
```

Logic Gates: OR

Find records meeting **at least one** criterion.

```
SELECT title FROM longlist
WHERE author = 'Fernanda Melchor' OR year = 2018;
```

Pattern Matching with LIKE

When you don't know the exact string.

- `%` : Any number of characters.
- `_` : Exactly one character.

LIKE Examples

```sql
-- Starts with 'The'
SELECT title FROM longlist WHERE title LIKE 'The%';

-- Ends with 'ed'
SELECT title FROM longlist WHERE title LIKE '%ed';

-- Contains 'night'
SELECT title FROM longlist WHERE title LIKE '%night%';
```

Case Sensitivity

- **PostgreSQL:** `LIKE` is case-sensitive (`ILIKE` for insensitive).
- **SQLite:** `LIKE` is case-insensitive for ASCII characters.

Dealing with Absence: NULL

In SQL, `NULL` is not `0`. It is not `""` (empty string).

It represents **Unknown** or **Missing** data.

6. Filtering NULLs

You cannot use `= NULL`. You must use `IS NULL`.

```sql
-- Books with no translator
SELECT title FROM longlist WHERE translator IS NULL;
```

IS NOT NULL

```sql
-- Only show translated works
SELECT title, translator FROM longlist
WHERE translator IS NOT NULL;
```

Range Filtering: BETWEEN

Inclusive range helper.

```
SELECT title, year FROM longlist
WHERE year BETWEEN 2010 AND 2020;
```

The IN Operator

Filtering against a set of values.

```
SELECT title FROM longlist
WHERE year IN (2019, 2021, 2023);
```

Sorting Data: ORDER BY

By default, databases return data in no guaranteed order (often insertion order).

```sql
SELECT title, year FROM longlist ORDER BY year;

SELECT title, year FROM longlist ORDER BY title;
```

ASC vs DESC

- **ASC:** Smallest to largest (Default).
- **DESC:** Largest to smallest.

```sql
SELECT title, rating FROM longlist ORDER BY rating DESC;
```

Multi-Level Sorting

Sort by year (newest first), then by title (alphabetically).

```
SELECT year, title FROM longlist
ORDER BY year DESC, title ASC;
```

Arithmetic in SELECT

SQL can perform calculations on the fly.

```
SELECT title, (votes / 100) AS popularity_index
FROM longlist;
```

Aliasing with AS

Rename columns in your output for clarity.

```
SELECT COUNT(*) AS total_books FROM longlist;
```

8. Aggregate Functions

Functions that take a collection of values and return a single scalar.

COUNT()

```sql
-- Total rows
SELECT COUNT(*) FROM longlist;

-- Total non-null translators
SELECT COUNT(translator) FROM longlist;
```

SUM() and AVG()

```sql
-- Average rating of all books
SELECT AVG(rating) FROM longlist;

-- Total votes cast in 2021
SELECT SUM(votes) FROM longlist WHERE year = 2021;
```

MIN() and MAX()

```sql
-- Find the lowest and highest rating
SELECT MIN(rating), MAX(rating) FROM longlist;
```

Formatting: ROUND()

Keep your output clean.

```
SELECT ROUND(AVG(rating), 2) AS 'Clean Average'
FROM longlist;
```

DISTINCT Values

Removing duplicates from your result set.

```sql
-- List unique years in the database
SELECT DISTINCT year FROM longlist ORDER BY year;
```

Nested Functions

You can combine aggregates and math.

```
SELECT MAX(rating) - MIN(rating) AS rating_spread
FROM longlist;
```

Comments in SQL

- `--` : Single line comment.
- `/* ... */` : Multi-line comment.

[PRACTICE] Mini-Challenge 1

Find the top 3 books from 2022 with the most votes.

- *Filter:* `year = 2022`
- *Sort:* `votes DESC`
- *Limit:* `3`

[PRACTICE] Mini-Challenge 2

Find the average rating for books that contain the word "Chronicle" in the title.

- *Filter:* `LIKE '%Chronicle%'`
- *Function:* `AVG(rating)`

Common Errors to Avoid

- Missing semicolons `;`
- Using `"` for strings (should be `'`)
- Filtering by aggregate functions in `WHERE` (requires `HAVING`, coming next week!).

SQL Execution Order

Even though you write `SELECT` first, the database executes:

1. **FROM** (Get data)
2. **WHERE** (Filter)
3. **SELECT** (Choose columns)
4. **ORDER BY** (Sort)
5. **LIMIT** (Clean up)

Summary: Today's Toolkit

- **Environment:** SQLite + VSC.
- **Retrieval:** SELECT, DISTINCT.
- **Refinement:** WHERE, LIKE, BETWEEN, IN, NULL.
- **Organization:** ORDER BY, LIMIT.
- **Math:** COUNT, SUM, AVG, MIN, MAX.

Practice Resource

Let's visit:

SQL Police Department

Solve crimes using your new SQL skills!

MTM4692 - Readiness Survey

https://docs.google.com/forms/d/e/1FAIpQLSeZktvCwdnPSH7dtbq_MKarB2qLdd7uT1FUzBIU4d-jBZDL-A/viewform?usp=publish-editor

Questions & Discussion

Thank you!

Happy Ramadan!